Logistic Regression Model for the Data of Evans

Group 5

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Introduction:

A linear regression model is applied when our response follows a normal distribution. In some practical applications, the response may be continuous, binary, count or so on. In those situations, we move to generalized linear models. In the generalized linear models, we can account for various types of response, such as, binary response, continuous response and so on by fitting the response to a desired distribution . When our response is binary, the best approach is to use the generalized linear model with the response following a binomial distribution, which is the logistic regression model.
See: Sharandeep Singh Pandher (Chapter 1. Introduction to multiple linear regression & Chapter 2. GLMs)

In the given project dataset, the data was collected from a cohort study in which 609 observations on 9 variables (CDH, CAT, AGE, CHL, SMK, ECG, DBP, SBP, and HPT) were collected from white males in Evans County who were followed for 7 years, with coronary heart disease as the outcome of interest. The goal of the study was to identify influential factors associated with coronary heart disease, that is, variable selection. Simultaneously, we aimed to select the best-fitted model based on the $\chi 2$ test of deviances for model selection.

See: D. Kleinbaum and M. Klein (2010) Survival Analysis: A Self-Learning Text. 3rd ed. Springer.

1.CDH : outcome variable; 1 = coronary heart disease

2.CAT : 1 = high, 0 = normal catecholamine level

3.AGE : age (in years)

4.CHL : cholesterol, mg/dl

5.SMK : 1 = subject has ever smoked

6.ECG : 1 = presence of electrocardiogram abnormality

7.DBP : diastolic blood pressure, mmHg

8.SBP : systolic blood pressure, mmHg

9.HPT : 1 = SBP greater than or equal to 160 or DBP greater than or equal to 95

Note: The data set ”Evans” is available in r-package ”lbreg”.

We will use binary logistic regression to develop a model that can estimate catecholamine level, age, cholesterol, whether they have ever smoked, presence of electrocardiogram abnormality, diastolic blood pressure, mmHg, systolic blood pressure, and whether they have HPT (SBP greater than or equal to 160 or DBP greater than or equal to 95)."

Methods:

Method for hypothesis tests:

Compare two nested models: When comparing two nested models, under the null hypothesis of no difference between the two models, the difference in the deviances of the two models asymptotically follows the χ^2 distribution, with degrees of freedom "d" being the difference of the number of parameters in the two models being compared, i.e.

$$D_R - D_F \rightarrow \chi^2(\mathrm{d}), \mathrm{n} \rightarrow \infty$$

where D_R and D_F are the deviances for the restricted model and the full model respectively. This result can be used for model comparison and model selection.

Method for finding over-dispersion:

For a logistic regression model, we may assume that

$$\mathrm{Var}(y_i) = \tau \mathrm{E}(y_i)(1 - \mathrm{E}(y_i))$$

where τ is called a dispersion parameter. If $\tau = 1$, then y_i follows a binomial distribution. If $\tau \neq 1$, then y_i does not follow a binomial distribution.

Data Analysis:

Fitting the Model

```
log = glm(CDH ~ ., family = binomial, data = evans)
summary(log)
```

```
>
> Call:
> glm(formula = CDH ~ ., family = binomial, data = evans)
>
> Deviance Residuals:
>     Min       1Q   Median       3Q      Max
> -1.1291  -0.5222  -0.4033  -0.2835   2.6444
>
> Coefficients:
>              Estimate Std. Error z value Pr(>|z|)
> (Intercept) -6.764709   1.579763  -4.282 1.85e-05 ***
> CATTRUE      0.750257   0.377231   1.989  0.04672 *
> AGE          0.037781   0.016031   2.357  0.01844 *
> CHL          0.008674   0.003296   2.632  0.00850 **
> SMKTRUE      0.867474   0.307218   2.824  0.00475 **
> ECGTRUE      0.403426   0.296125   1.362  0.17309
> DBP          0.012402   0.014284   0.868  0.38526
> SBP         -0.010571   0.008073  -1.309  0.19038
> HPTTRUE      0.532752   0.383747   1.388  0.16505
```

```
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> (Dispersion parameter for binomial family taken to be 1)
>
>     Null deviance: 438.56  on 608  degrees of freedom
> Residual deviance: 398.60  on 600  degrees of freedom
> AIC: 416.6
>
> Number of Fisher Scoring iterations: 5
```

Logistic Regression model is used over a MLR model to capture the nature of the response (binary) by fitting it to a binomial distribution.

Compared with linear regression, the logistic regression model is a valuable tool for modeling and predicting binary outcomes. Hence, we consider the logistic regression model as opposed to the usual linear regression model to describe the relationship.

Over-dispersion

```
DIS = sum(residuals(log, type="pearson")^2)/log$df.res
print(DIS)
```

```
> [1] 1.002793
```

DIS is roughly equal to 1, hence no over-dispersion problem exists, suggesting that the results from the model log may be reliable.

Variable Selection Using Stepwise AIC Values

```
log2 = stepAIC(log, direction = c("both"), k = 2)

summary(log2)
```

```
>
> Call:
> glm(formula = CDH ~ CAT + AGE + CHL + SMK + HPT, family = binomial,
>     data = evans)
>
> Deviance Residuals:
>     Min       1Q   Median       3Q      Max
> -1.0762  -0.5297  -0.3995  -0.3048   2.5356
>
> Coefficients:
>              Estimate Std. Error z value Pr(>|z|)
> (Intercept) -6.680112   1.136363  -5.879 4.14e-09 ***
> CATTRUE      0.715810   0.340180   2.104  0.03536 *
> AGE          0.032770   0.015197   2.156  0.03105 *
> CHL          0.008608   0.003259   2.641  0.00827 **
> SMKTRUE      0.802906   0.303001   2.650  0.00805 **
> HPTTRUE      0.476272   0.289296   1.646  0.09970 .
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
>
> (Dispersion parameter for binomial family taken to be 1)
>
>     Null deviance: 438.56  on 608  degrees of freedom
> Residual deviance: 401.95  on 603  degrees of freedom
> AIC: 413.95
>
> Number of Fisher Scoring iterations: 5
```

ECG, DBP, SBP are removed from the original model based on stepwise AIC values.

Checking for Co-variate Significance by Comparing Step-wise Nested Models Using Chi-Square Hypothesis Tests

Wald-type test is not used due to the presence of categorical variables. Instead we can compare models by dropping/adding one covariate at a time using a χ^2 test.

```
anova(log2, test="Chi")
```

```
> Analysis of Deviance Table
>
> Model: binomial, link: logit
>
> Response: CDH
>
> Terms added sequentially (first to last)
>
>
>      Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
> NULL                   608     438.56
> CAT   1  14.1312       607     424.43 0.0001705 ***
> AGE   1   4.0483       606     420.38 0.0442171 *
> CHL   1   8.2166       605     412.16 0.0041510 **
> SMK   1   7.5056       604     404.66 0.0061506 **
> HPT   1   2.7097       603     401.95 0.0997410 .
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

All covariates are significant in log2 at a 0.1 significance level. (HPT not significant at 0.05 level).

New model without HPT:

```
log3 = glm(CDH~CAT+ AGE+ CHL+ SMK, family = binomial, data = evans)

summary(log3)
```

```
>
> Call:
> glm(formula = CDH ~ CAT + AGE + CHL + SMK, family = binomial,
>     data = evans)
```

```
>
> Deviance Residuals:
>     Min       1Q   Median       3Q      Max
> -1.0562  -0.5230  -0.4064  -0.3150   2.4855
>
> Coefficients:
>              Estimate Std. Error z value Pr(>|z|)
> (Intercept) -6.659008   1.127218  -5.907 3.47e-09 ***
> CATTRUE      0.931183   0.315154   2.955  0.00313 **
> AGE          0.033141   0.015171   2.184  0.02893 *
> CHL          0.009294   0.003228   2.879  0.00398 **
> SMKTRUE      0.789630   0.301677   2.617  0.00886 **
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> (Dispersion parameter for binomial family taken to be 1)
>
>     Null deviance: 438.56  on 608  degrees of freedom
> Residual deviance: 404.66  on 604  degrees of freedom
> AIC: 414.66
>
> Number of Fisher Scoring iterations: 5
```

Model Selection by Comparing the Nested Models (full vs restricted) using ANOVA

We set up the hypothesis test as:

H_0: There are no differences between the models log2 and log.

H_A: There are some differences between the models log2 and log.

```
anova(log2, log)
```

```
> Analysis of Deviance Table
>
> Model 1: CDH ~ CAT + AGE + CHL + SMK + HPT
> Model 2: CDH ~ CAT + AGE + CHL + SMK + ECG + DBP + SBP + HPT
>   Resid. Df Resid. Dev Df Deviance
> 1       603     401.95
> 2       600     398.60  3   3.3468
```

We see that the difference in deviances is 3.3468, and models differ by 3 degrees of freedom.

The p-value of a chi square distribution with 3 degrees of freedom at 3.3468 is 0.34117, that is, $P(\chi^2_3 \geq 3.3468) = 0.34117$

At a 5% significance level, since p-value > α = 0.05, we do not reject the null hypothesis, that is, there are no significant differences among the models, hence we should choose the smaller model which is log2.

Model Selection by Comparing the Restricted Nested Models using ANOVA

We set up the hypothesis test as:

H_0: There are no differences between the models log3 and log2.

H_A: There are some differences between the models log3 and log2.

```
anova(log3, log2)
```

```
> Analysis of Deviance Table
>
> Model 1: CDH ~ CAT + AGE + CHL + SMK
> Model 2: CDH ~ CAT + AGE + CHL + SMK + HPT
>   Resid. Df Resid. Dev Df Deviance
> 1       604     404.66
> 2       603     401.95  1   2.7097
```

We see that the difference in deviances is 2.7097, and models differ by 1 degree of freedom.

The p-value of the chi square distribution with 1 degree of freedom at 2.7097 is 0.09977, that is, $P(\chi^2_1 \geq 2.7097) = 0.09977$

At a 5% significance level, since p-value > α = 0.05, we do not reject the null hypothesis, that is, there are no significant differences among the models, hence we should choose the smaller model which is log3.

Conclusion:

In the project, the response variable CDH is binary, taking only two possible values (say, 0 or 1). Predictors are CAT, AGE, CHL, SMK, ECG, DBP, SBP, and HPT.

In this case, we assume that the response variable follows a binomial distribution. We fit binary logistic regression as a full model, and find no over-dispersion problem exists.

Using stepAIC() command in R, we identify the most important predictors for predicting the response variable are CAT, AGE, CHL, SMK, HPT, that is, influential factors associated with coronary heart disease are CAT, AGE, CHL, SMK, HPT.

Then we use anova() to test each covariate sequentially. The result is that significant covariates are CAT, AGE, CHL and SMK. Finally, we get one full model(log1), one restricted model(log2) without predictors ECG, DBP, SBP and one restricted model(log3) without predictors ECG, DBP, SBP, HPT. By comparing them respectively as nested models, our model selection is log3.

Appendix:

```
## Initialization
rm(list=ls())
library(lbreg)
library(MASS)

## Accessing dataset
data(Evans)
head(Evans)

## Initializing Response and Covariates
CDH = Evans[,1]

CAT = Evans[,2]
AGE = Evans[,3]
CHL = Evans[,4]
SMK = Evans[,5]
ECG = Evans[,6]
DBP = Evans[,7]
SBP = Evans[,8]
HPT = Evans[,9]

## Factoring to encode categorical variables
help(Evans)
evans <- with(Evans, {
  data.frame(CAT = (CAT > 0), AGE, CHL, SMK = (SMK >0), ECG = (ECG>0), DBP,
SBP, HPT = (HPT>0))
})

# Question 1
## Logistic Regression model on dataset
log = glm(CDH ~ ., family = binomial, data = evans)
summary(log)

## Logistic Regression model is used over a MLR model
## to capture nature of the response (binary) by fitting it to a binomial
distribution

## Over-dispersion problem (DIS roughly 1 so no overdispersion problem)
DIS = sum(residuals(log, type="pearson")^2)/log$df.res
print(DIS)

# Question 2
```

```r
## Variable selection using AIC
log2 = stepAIC(log, direction = c("both"), k = 2)
summary(log2)

## ECG, DBP, SBP are removed based on stepwise AIC values
## Checking significance of each covariate sequentially using chi-square
hypothesis test
anova(log2, test="Chi")

## All covariates are significant in log2 (HPT not significant at 0.05 level)
## New model without HPT
log3 = glm(CDH~CAT+ AGE+ CHL+ SMK, family = binomial, data = evans)
summary(log3)


## Comparing the nested models (full vs restricted) using ANOVA
anova(log2, log)

## Wald-type test is not used due to presence of categorical variables
## Difference in deviances is 3.3468, and models differ by 3 degrees of
freedom
## p-value of chi square is 0.34117
## since p-value > 0.05, we do not reject null hypothesis
## hence choose smaller model which is log2

## Comparing the restricted nested models using ANOVA
anova(log3, log2)

## Difference in deviances is 2.7097, and models differ by 1 degree of
freedom
## p-value of chi square is 0.09977
## since p-value > 0.05, we do not reject null hypothesis
## we choose smaller model which is log3
```